

# Melanoma

Dr eman gomaa

Consultant dermatologist

SCE Dermatology / EBDV by dr :Eman Gomaa - CoNNect Academy

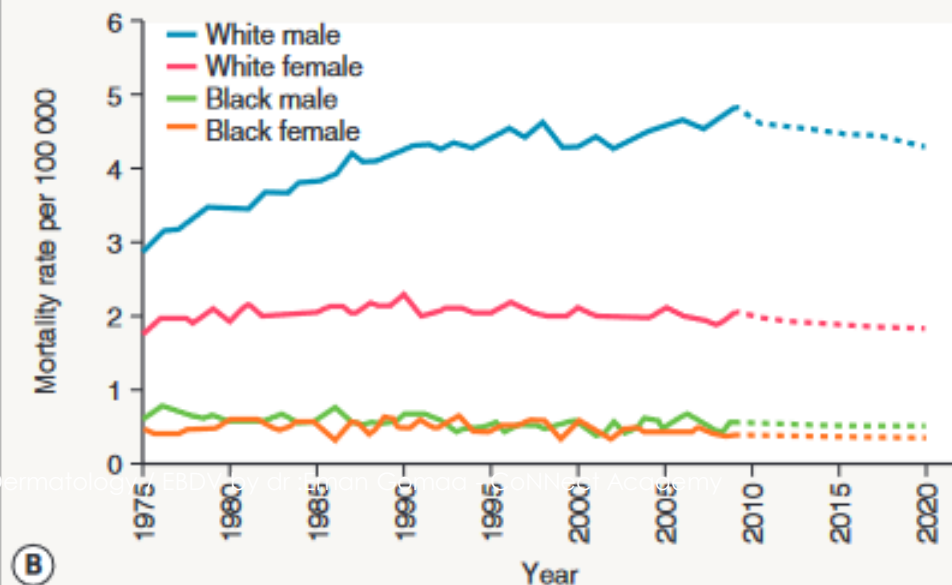
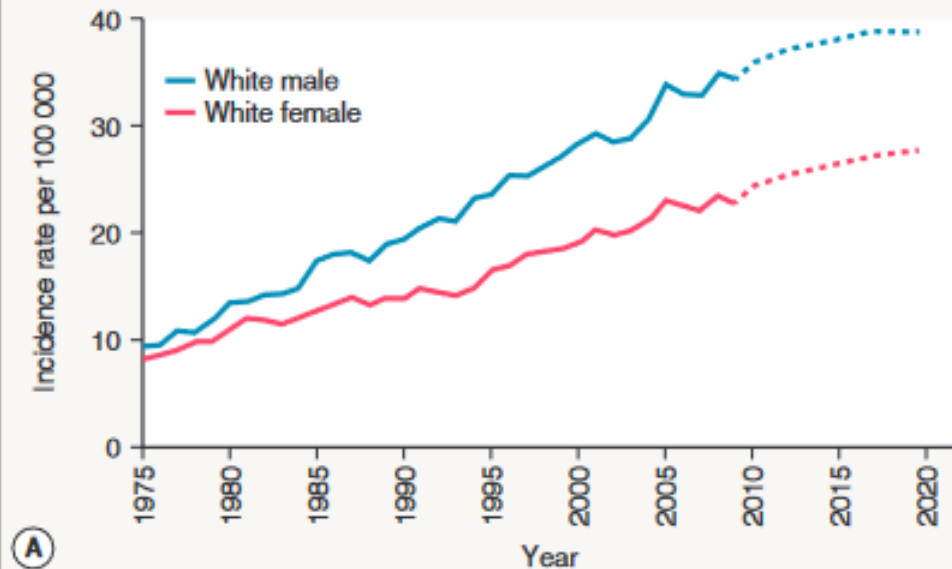
# melanoma

- Malignant tumor of melanocyte
- Most common cutaneous origen
- Mucosal surface , uveal tract
- leptomeninges



- Caucasian white skin , Australia , newzeland
- Increase world wide
- 5-7 folds in 4<sup>th</sup> decade
- Median age 60yyears

**INCIDENCE RATES AND MORTALITY RATES FOR MELANOMA IN THE US PER 100 000 INHABITANTS (1975–2010, PROJECTED TO 2020)**



**Fig. 113.4** Incidence rates and mortality rates for melanoma in the US per 100 000 inhabitants (1975–2010, projected to 2020). **A,B** While the incidence has increased significantly in white men and women over the past four decades, there has been a modest increase in mortality in white men. *Data from [www.cdc.gov/cancer/dcpc/research/articles/cancer\\_2020\\_incidence.htm](http://www.cdc.gov/cancer/dcpc/research/articles/cancer_2020_incidence.htm) or [figures.htm](http://www.cdc.gov/cancer/dcpc/research/articles/cancer_2020_figures.htm).*

# aetiopathogenesis



## Genetic

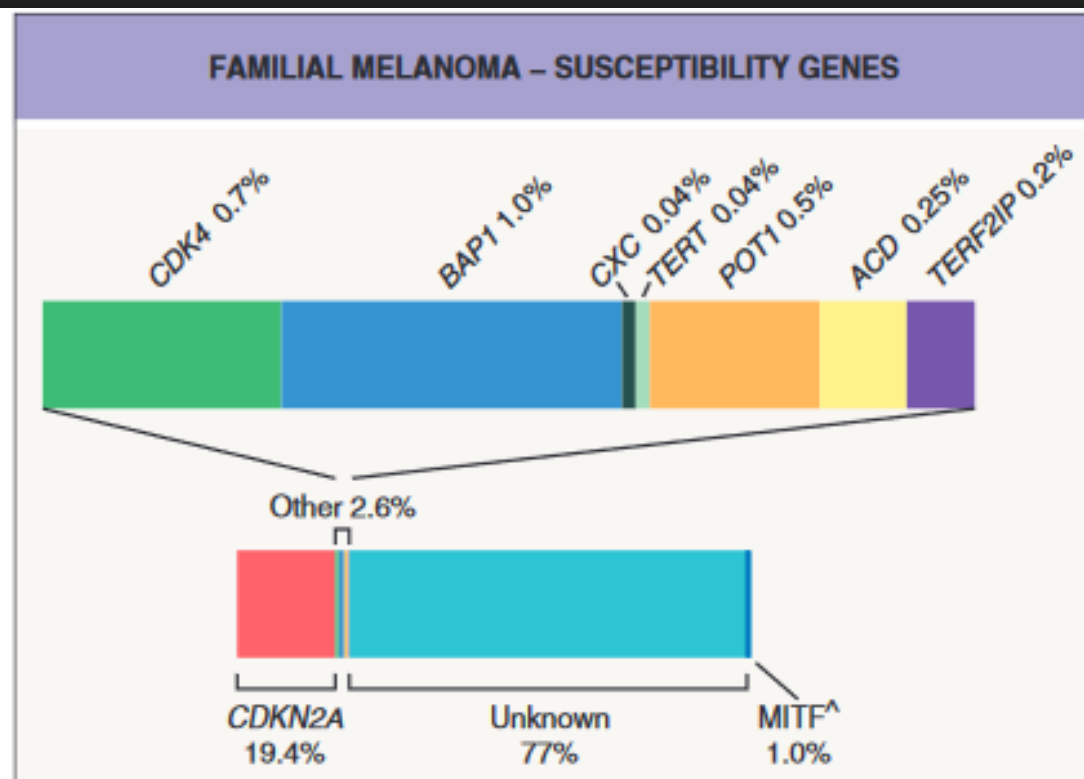
Family history  
1,3-15,8%  
2folds increase risk of  
melanoma

Gene mutaton  
CDKN2A  
CDK4

DNA repair defect XP

## Phenotypic triat

Fair skin  
Tendacy to burn  
Inability to tan  
Red hair



**Fig. 113.5 Familial melanoma – susceptibility genes.** Germline *CDKN2A* mutations are observed in ~20% of familial melanoma kindreds. Note the significant percentage of families for whom the genetic basis is unknown, a point that should be discussed with patients contemplating genetic screening. Patients with mutations in *CDKN2A* (p16), *CDKN2A* (p14), *MITF*, and *POT1* can also develop pancreatic carcinoma, neural tumors, renal cell carcinoma, and gliomas, respectively, while those with *BAP1* mutations can develop uveal melanoma, mesothelioma, renal cell carcinoma, cholangiocarcinoma, and basal cell carcinoma (at an early age). ACD\*, adrenocortical dysplasia protein homolog; BAP1, BRCA1 associated protein 1; CDK4, cyclin-dependent kinase 4; CXC, CXC motif chemokine; MITF, microphthalmia-associated transcription factor; POT1\*, protection of telomeres 1; TERF2IP\*, telomeric repeat binding factor 2 Interacting protein; TERT, telomerase reverse transcriptase (mutations in promoter). \*Component of shelterin (see Fig. 67.22). ^Population attributable fraction. Adapted from Potrony M, Badenas C, Aguilera P, et al. Update in genetic

## RISK FACTORS FOR THE DEVELOPMENT OF CUTANEOUS MELANOMA

### Genetic factors

- Family history of cutaneous melanoma
- Lightly pigmented skin
- Tendency to burn, inability to tan
- Red hair color
- DNA repair defects (e.g. xeroderma pigmentosum)

### Environmental factors

- Intense Intermittent sun exposure
- Chronic sun exposure
- Residence in equatorial latitudes
- PUVA (possible)
- Tanning bed use, especially under the age of 35 years
- Iatrogenic or acquired immunosuppression

### Phenotypic expressions of gene/environment interactions

- Melanocytic nevi and solar lentigines:
    - Increased total number of acquired melanocytic nevi (MN)  
>100 MN, relative risk ~ 8–10-fold increased
    - Atypical melanocytic nevi (AMN)  
> 5 AMN, relative risk ~ 4–6-fold increased
    - Multiple solar lentigines (SL)  
Multiple SL, relative risk ~3–4-fold increased
- Relative risks (RR) are multiplicative: e.g. a person with >100 MN + >5 AMN + multiple SL has a relative risk ~  $10 \times 5 \times 3 = 150$ -fold
- Personal history of cutaneous melanoma



# Host immune response to melanoma

○ Is Melanoma immunogenic tumor ?

**CD8**

CTLA4

PD1  
(CD279)

CD28

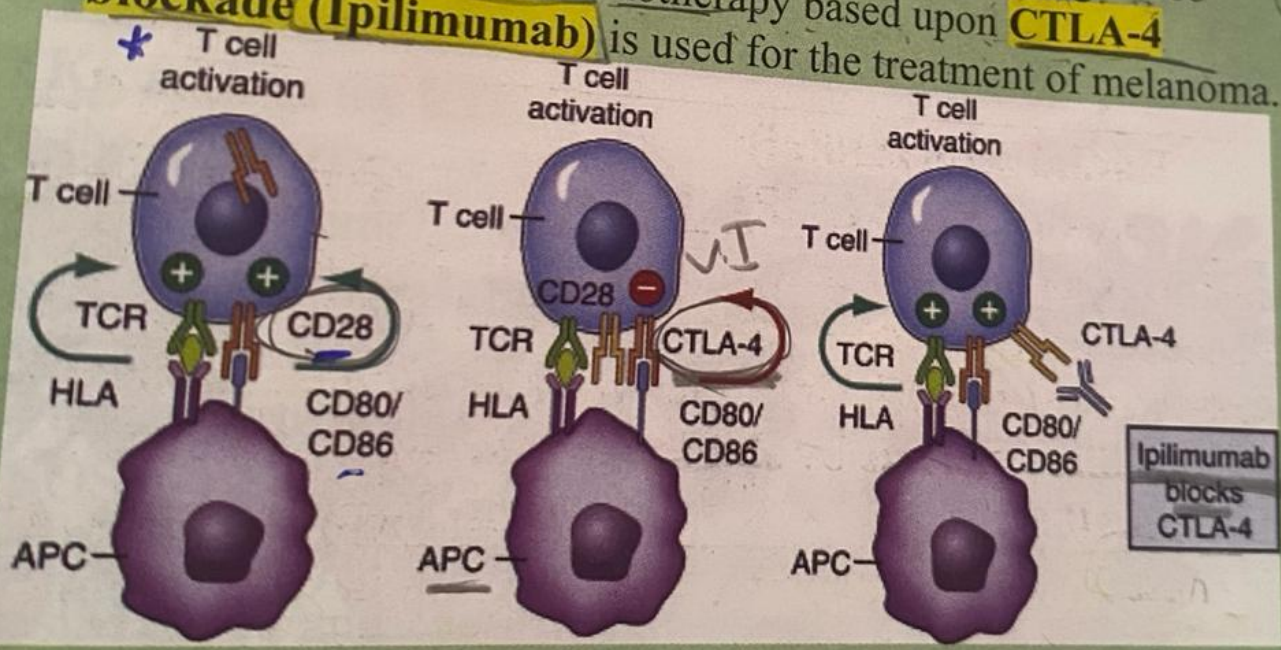
Anti CTLA4  
IPILIMUMAB

Anti PD1  
Nivolumab  
pembrolizumab

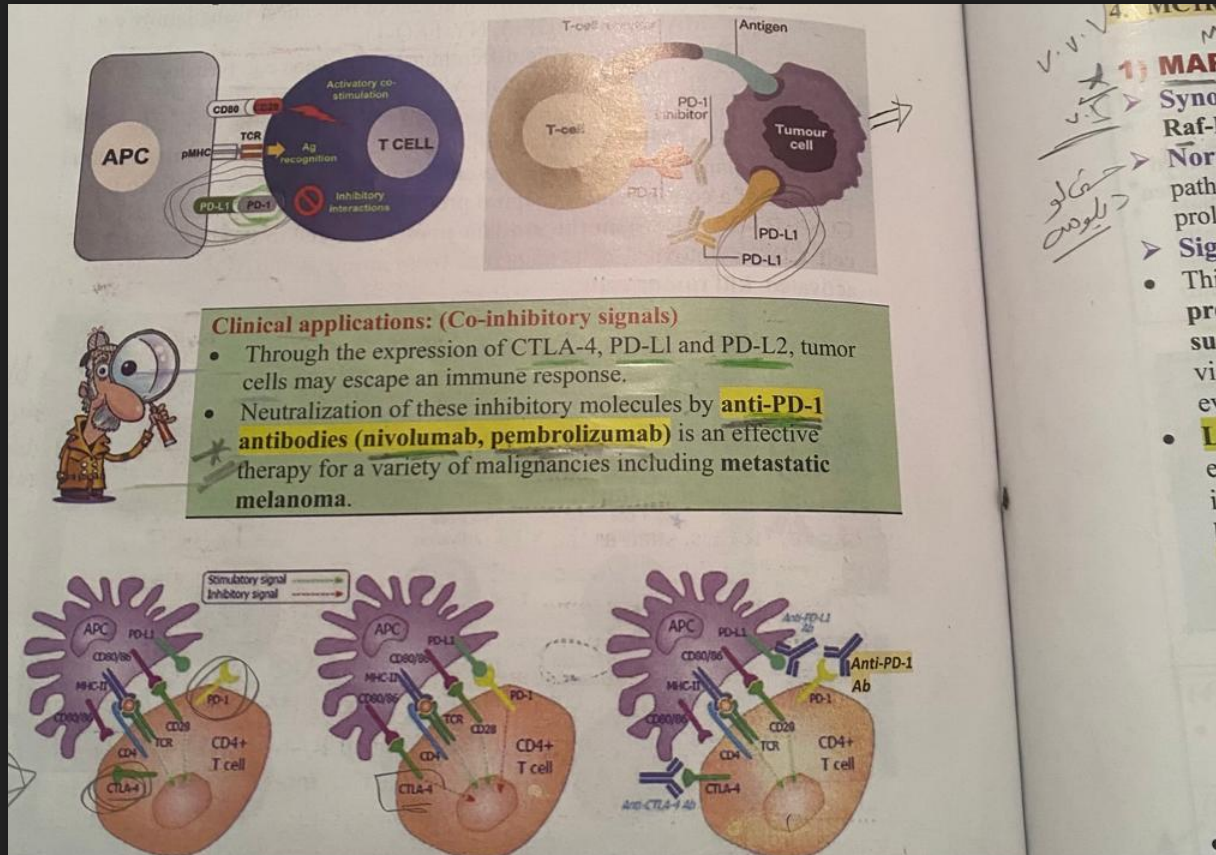
tumor cells. recognize these antigens and, if appropriately

**Clinical applications: (Co-inhibitory signals)**

- Cytotoxic T lymphocyte-associated antigen-4 (**CTLA-4** = **CD152**) is an **inhibitory** homologue of **CD28** that is up-regulated following T-cell activation and acts as a **natural inhibitor "brake"** of T-cell activation by removing the co-stimulatory signal. Immunotherapy based upon **CTLA-4 blockade (Ipilimumab)** is used for the treatment of melanoma.



Understanding immune mechanisms → development of melanoma vaccines

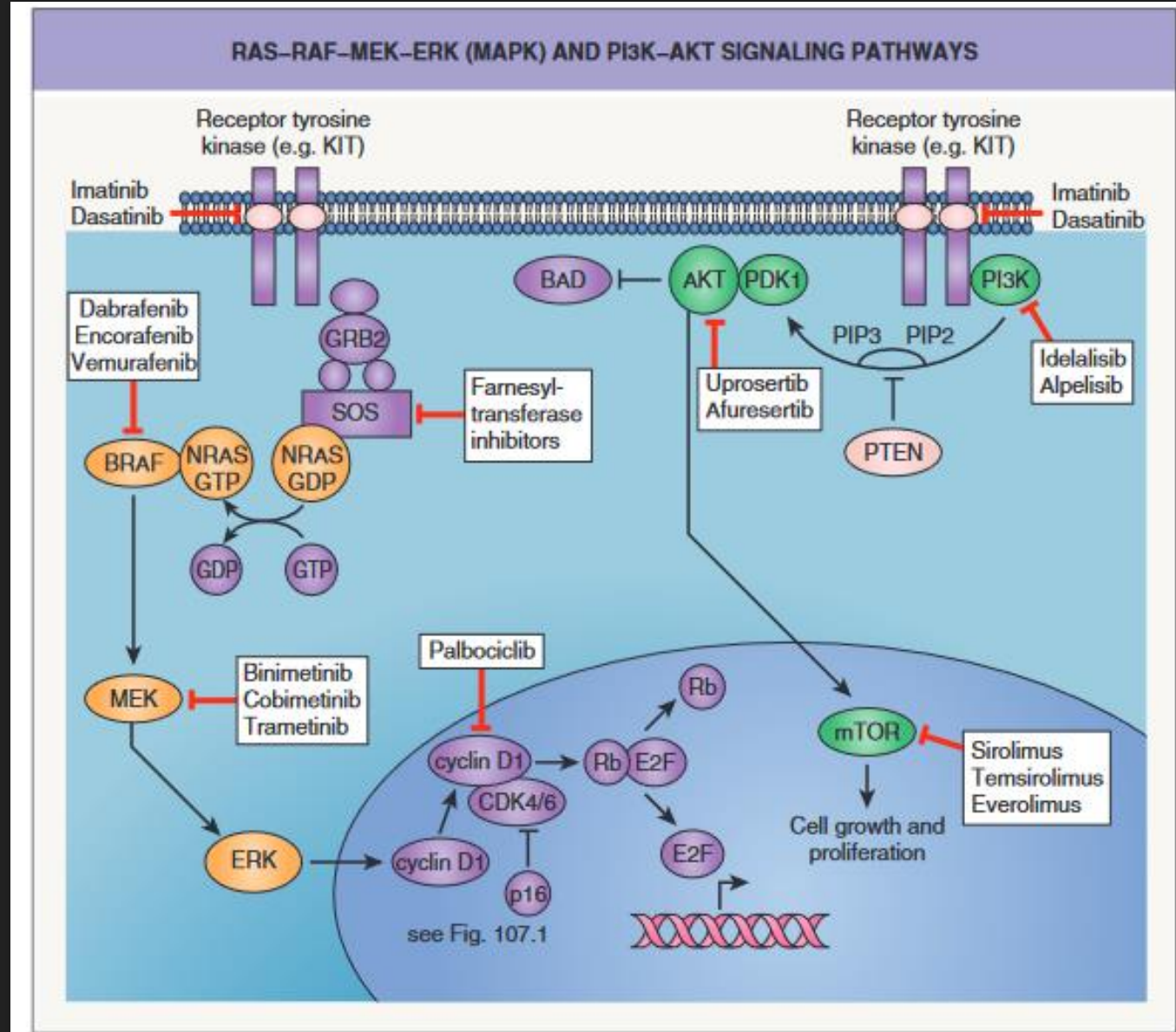


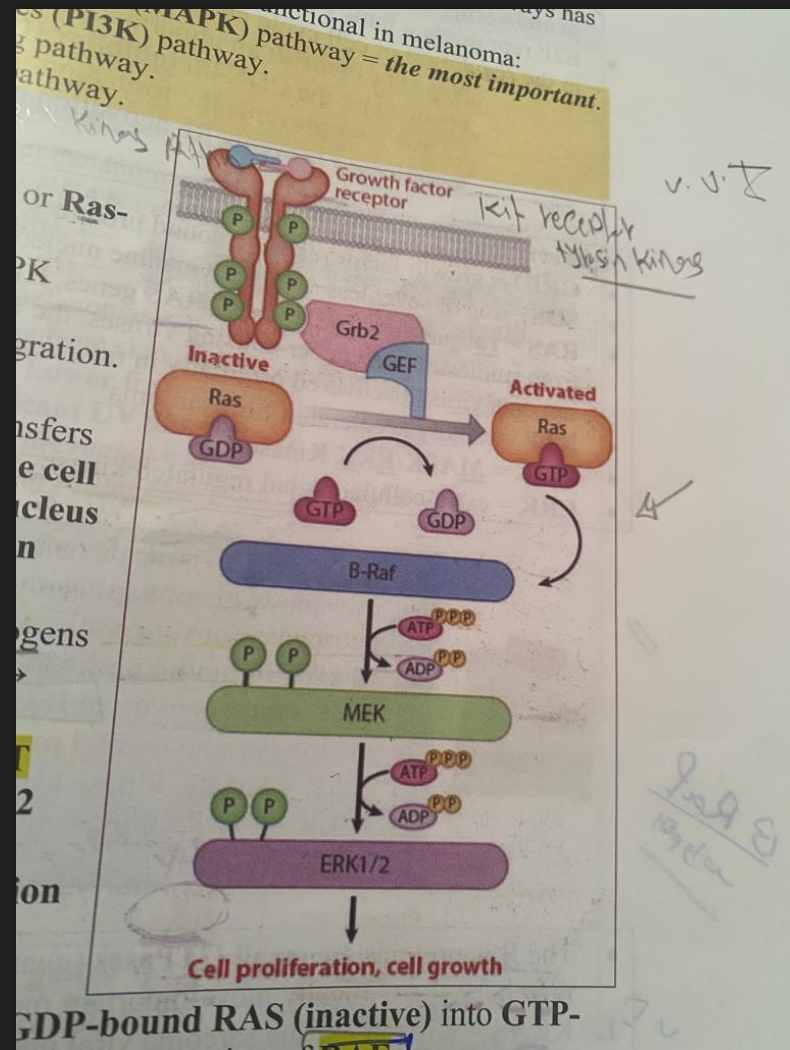
V.V.V.V. 1) MAPK  
 Synon  
 Raf-M  
 Norm  
 pathw  
 prolif  
 ➤ Sign  
 • This  
 prol  
 surf  
 via  
 eve  
 • Lig  
 e.g  
 int  
 ki  
 re  
 R  
 g  
 C



# Cellular signaling pathway

## MAPK PATHWAY



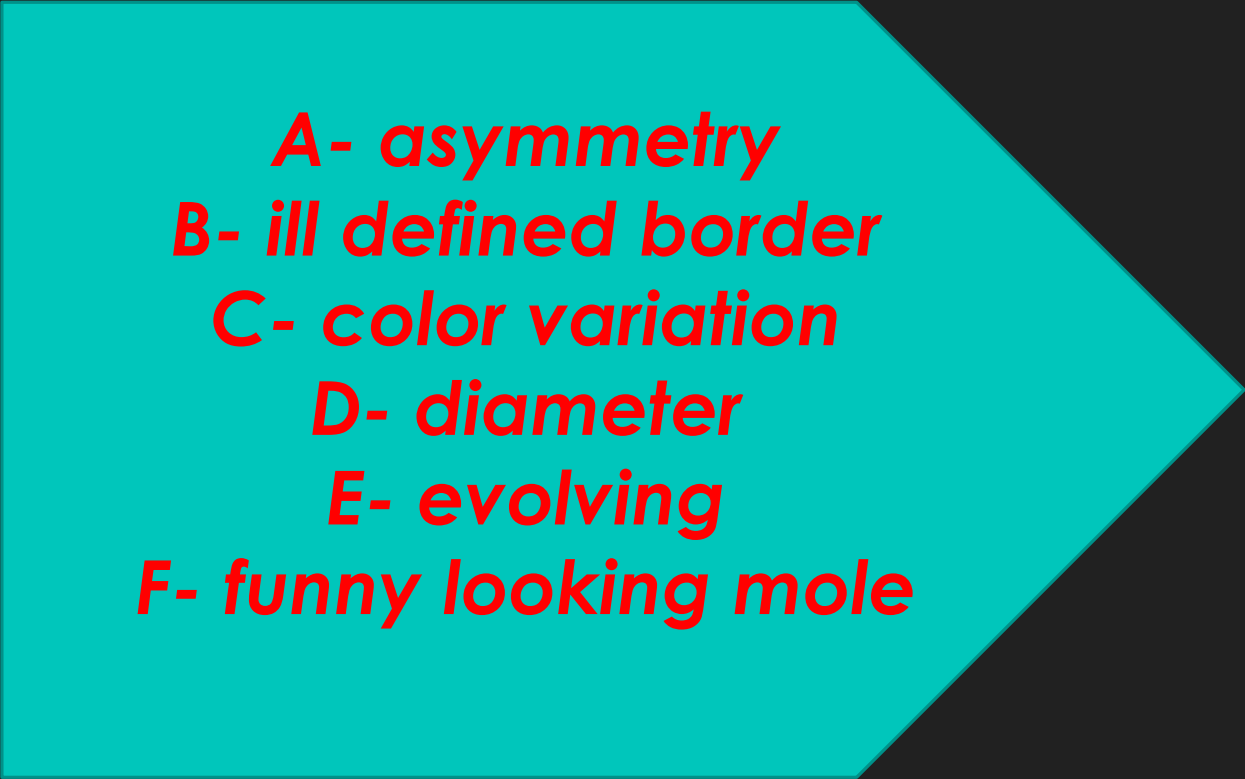


# Other pathways??

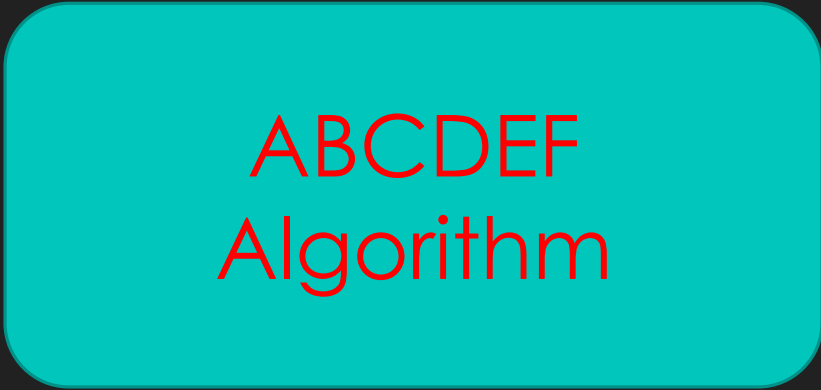
# Clinical picture

- Melanoma may arise denovo
- Or pre-existing melanocytic nevuse



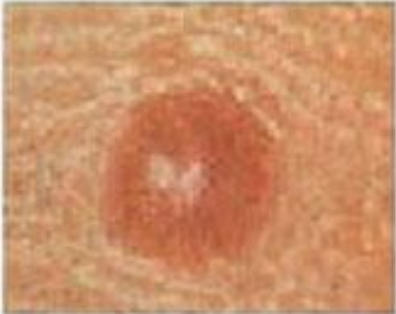


**A- asymmetry**  
**B- ill defined border**  
**C- color variation**  
**D- diameter**  
**E- evolving**  
**F- funny looking mole**



ABCDEF  
Algorithm

## Early Detection of Melanoma

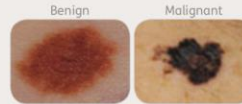
|          | Asymmetry                                                                          | Border                                                                              | Color                                                                                | Diameter                                                                             |
|----------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Normal   |   |   |   |   |
|          | Symmetrical                                                                        | Even Borders                                                                        | Single Color                                                                         | Smaller than 1/4"                                                                    |
| Melanoma |  |  |  |  |
|          | Asymmetrical                                                                       | Uneven Borders                                                                      | Multiple Colors                                                                      | Greater than 1/4"                                                                    |

# Melanoma check for the signs

Because melanoma is particularly serious, you should be familiar with the signs to look for. **The ABCDE of melanoma can help you to detect it early.**

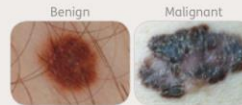
**A**

Is the spot asymmetric?



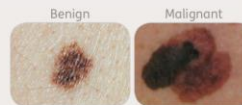
**B**

Does it have uneven borders?



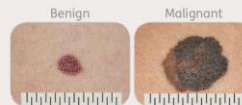
**C**

Does it contain different colours?



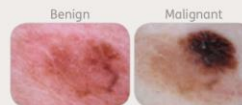
**D**

Is it larger than 6mm in diameter?



**E**

Is there an evolution in growth?



**EVOLUTION is the most important of the ABCDE warning signs.**

A spot that has changed in size, shape, colour, elevation or another trait (such as itching, soreness, bleeding or crusting).

MPA encourages everyone to get regular skin checks with a healthcare professional. If you notice new or changing spots MPA advises you to seek immediate medical advice.



Find out more and join the Melanoma Movement today  
[melanomapatients.org.au](http://melanomapatients.org.au)  
National Melanoma Support Line: 1300 88 44 50



| DIFFERENT TYPES OF PRIMARY CUTANEOUS MELANOMA |              |            |            |
|-----------------------------------------------|--------------|------------|------------|
| Clinico-histopathologic subtype               | Abbreviation | Percentage | Median age |
| Superficial spreading melanoma                | SSM          | 57.4%      | 51 years   |
| Nodular melanoma                              | NM           | 21.4%      | 56 years   |
| Lentigo maligna melanoma                      | LMM          | 8.8%       | 68 years   |
| Acral lentiginous melanoma                    | ALM          | 4%         | 63 years   |
| Unclassifiable melanoma                       | UCM          | 3.5%       | 54 years   |
| Others                                        |              | 5%         | 54 years   |

# Four major clinical variants

|  | Superficial spreading melanoma | Nodular melanoma                      | Lentigomaligna          | Acral lentiginous melanoma |
|--|--------------------------------|---------------------------------------|-------------------------|----------------------------|
|  | Most common<br>60-70 %         | 2 <sup>nd</sup> most common<br>15-30% | Minor case 10%          | Uncommon 5%                |
|  | 40-60 years                    | 6 <sup>th</sup> decade                | 7 <sup>th</sup> decade  | 7 <sup>th</sup> decade     |
|  | Men :trunk<br>Women :leg       | Trunk , head , neck                   | Face , nose ,<br>cheeks | Palm , sole , nail         |
|  |                                |                                       |                         |                            |

|                                                            | brown black macule ,<br>ABCDEF    | Blue black nodule<br>Rapidly developed<br>Ulceration | Slowly growing macule                            | Longitudinal melanonychia<br>Hutchinson sign<br>Extend to hyponychium |
|------------------------------------------------------------|-----------------------------------|------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------|
|                                                            | Slow radial growth<br>>>>vertical | Denovo vertical growth phase<br>>>without horizontal | Radial growth<br>5%progress to invasive melanoma |                                                                       |
|                                                            | 2 third regression                | Poor prognosis                                       | 5%progress to invasive melanoma                  | Very bad prognosis                                                    |
| SCE Dermatology / EBDV by dr :Eman Gomaa - CoNnect Academy |                                   |                                                      |                                                  |                                                                       |

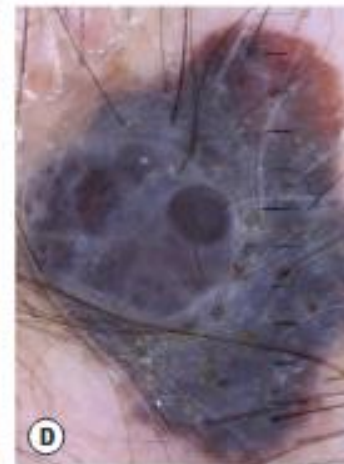


**Fig. 113.8** Longstanding superficial spreading melanoma. Note asymmetry, irregular borders, variation in colors, scar-like regression zones, and an inferior pink papule, indicating vertical growth phase.



**Fig. 113.10** Superficial spreading melanoma arising in a pre-existing nevus. Note the irregular outline and variable pigmentation.

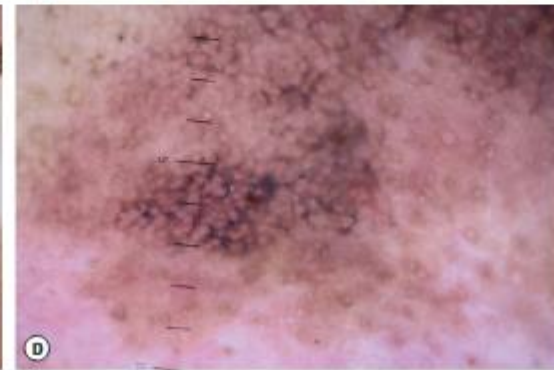
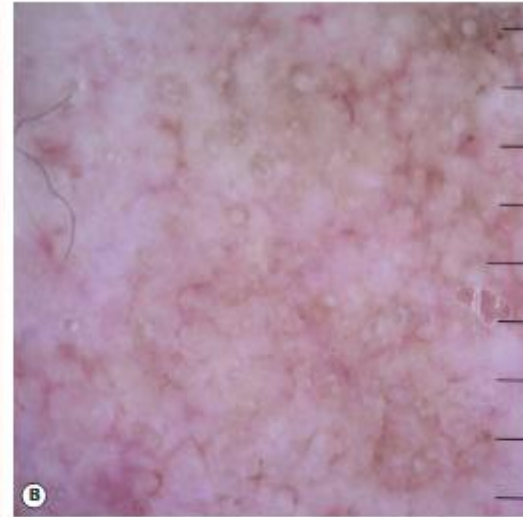




**Fig. 113.13 Nodular melanomas.** **A** Darkly pigmented papule measuring ~3 mm in diameter with a Breslow depth of 0.95 mm. **B** By dermoscopy, a blue color, atypical pigment network, and streaks are seen. **C** Darkly pigmented plaque on the scalp with an eccentric nodule; Breslow depth was 2.75 mm. **D** By dermoscopy, multiple colors including blue-gray are seen.



**Fig. 113.14** Lentigo maligna melanoma. **A** Asymmetric pigmented lesion on the earlobe, with irregular borders and significant variation in pigmentation; the Breslow depth was 0.45 mm. **B** Larger triangular brown patch in which there are multiple dark brown, black, and blue-gray papules and plaques. Breslow depth was 1.1 mm. (Courtesy: Kolman Marley MD)



**Fig. 113.15** Lentigo maligna. **A,C** An early lentigo maligna presenting as a light brown patch with subtle, asymmetric pigmentation is compared to a more advanced lesion with readily apparent variation in pigmentation. **B,D** By dermoscopy, follicular openings surrounded by annular pigmentation are seen, referred to as a "circle in a circle". Findings are more subtle in **B** and a few rhomboidal structures are seen in **D**.



**Fig. 113.16** Acral lentiginous melanoma. Irregularly pigmented lesion on the plantar surface of the foot. Dermoscopic findings include irregular diffuse pigmentation and a parallel ridge pattern (see Fig. 112.13).



**Fig. 113.17** Melanoma of the nail matrix. **A,B** The longitudinal pigmentation of the nail plate is nonhomogeneous, both clinically and dermoscopically. Histologically, the Breslow depth was 0.7 mm.

## Melanoma in pregnancy

Melanoma increase during pregnancy due to hormonal changes

Rare transplacental metastasis

Ultrasound , MRI

Noct , xray

Surgical excision with sentinel L.N piobsy

## Childhood melanoma

V RARE

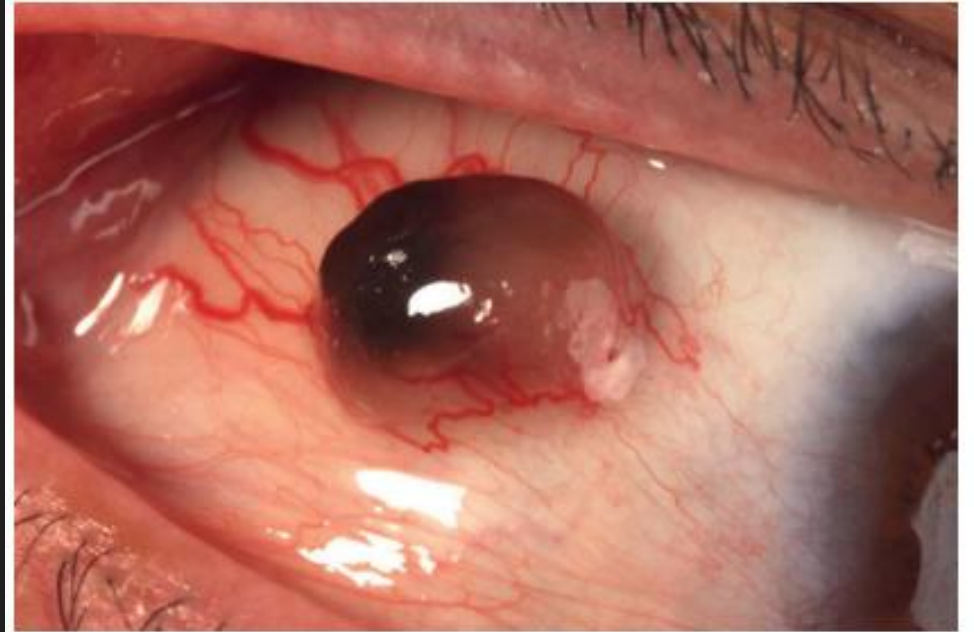
Xp

Gaint congenital melanocytic nevi

Lack of ABCD



OTHERS  
AMELANOTIC  
MUCOSAL  
OCULAR



eFig. 113.9 Conjunctival melanoma.



# Dermoscopic criteria

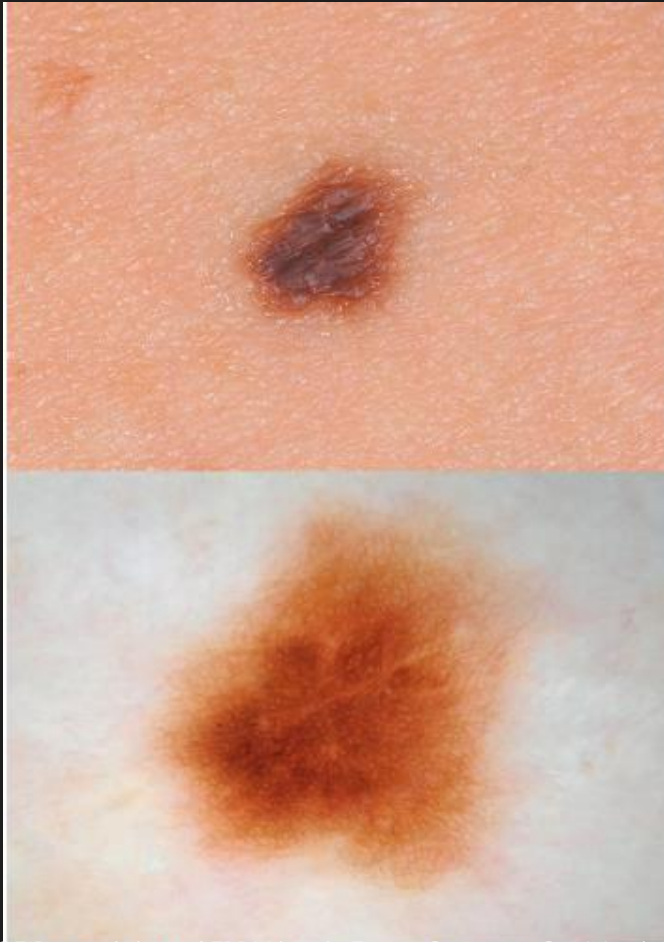


## 3 major

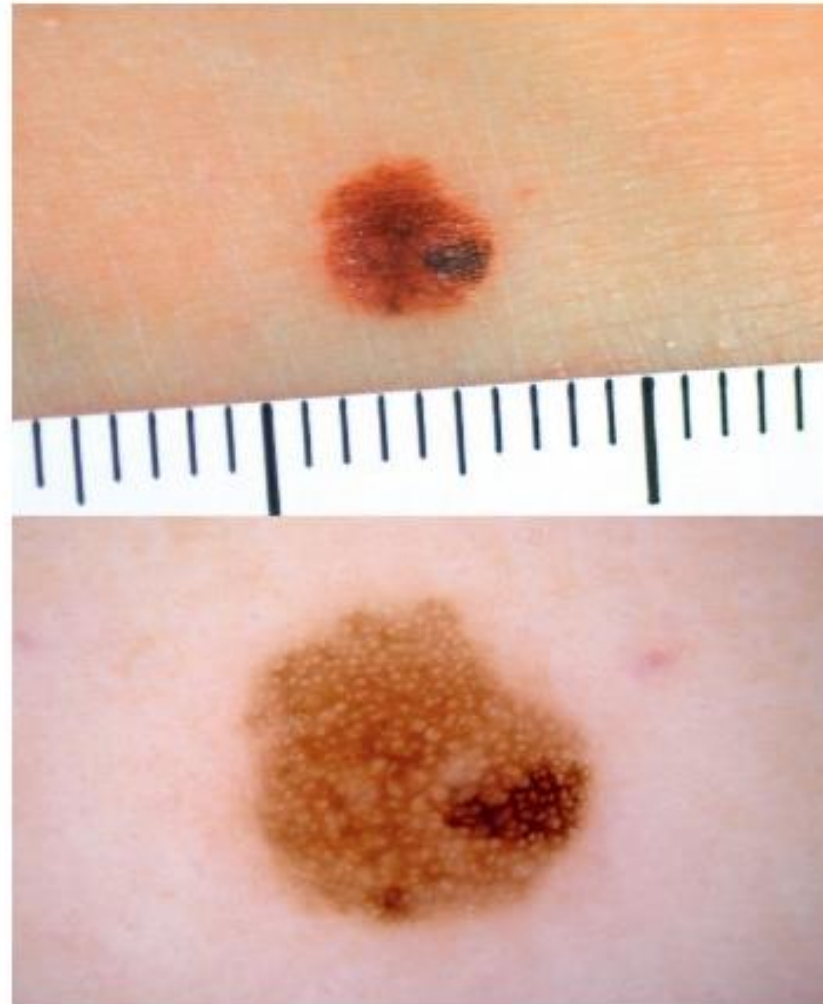
1. Atypical network
2. Atypical vascular pattern
3. Blue white veil

## 4 minor

1. Atypical dots , globules
2. Atypical blotches
3. Atypical streaks
4. Regression structure

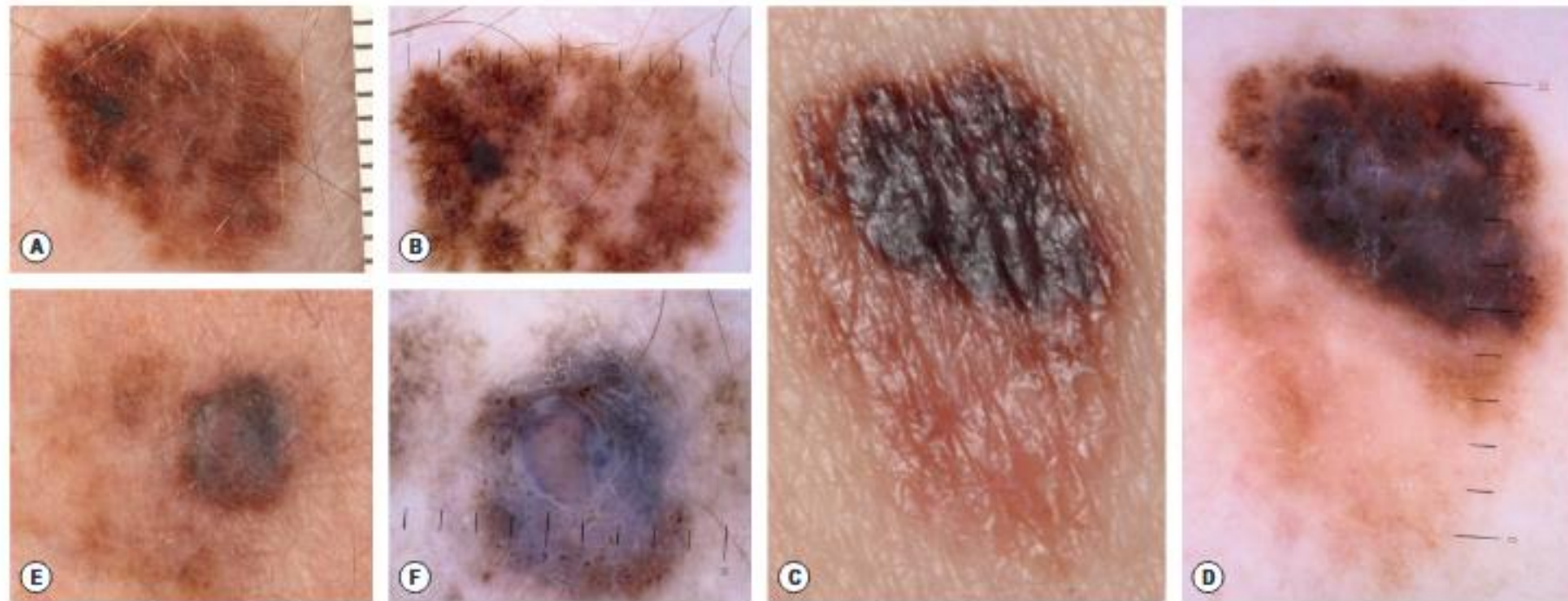


SC **Fig. P2 Junctional nevus.** Clinically, a brown macule with central hyperpigmentation. Dermoscopically, a uniform pigment network.



**Fig. P5 Clinically atypical melanocytic nevus.** The dermoscopy pattern is reticular-disorganized and can be seen with uncertain lesions.

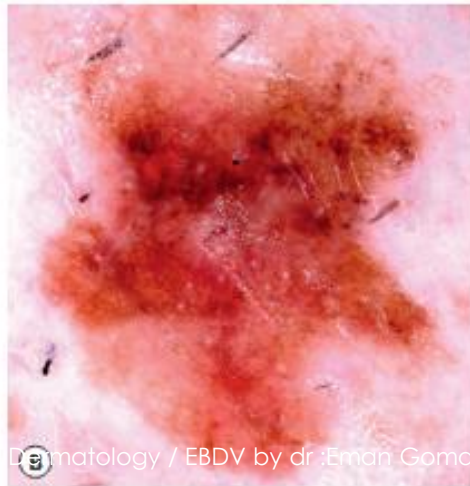




**Fig. 113.9 Superficial spreading melanomas.** Clinically, all of these lesions demonstrate asymmetry due to variation in color and irregularity in outline. Breslow depths for A, C, E were <0.5 mm, 0.58 mm, and 1.60 mm, respectively. B,D,F By dermoscopy, there is asymmetry, atypical pigment networks, irregular blotches, and multiple colors; in D and F a blue-white veil is present.

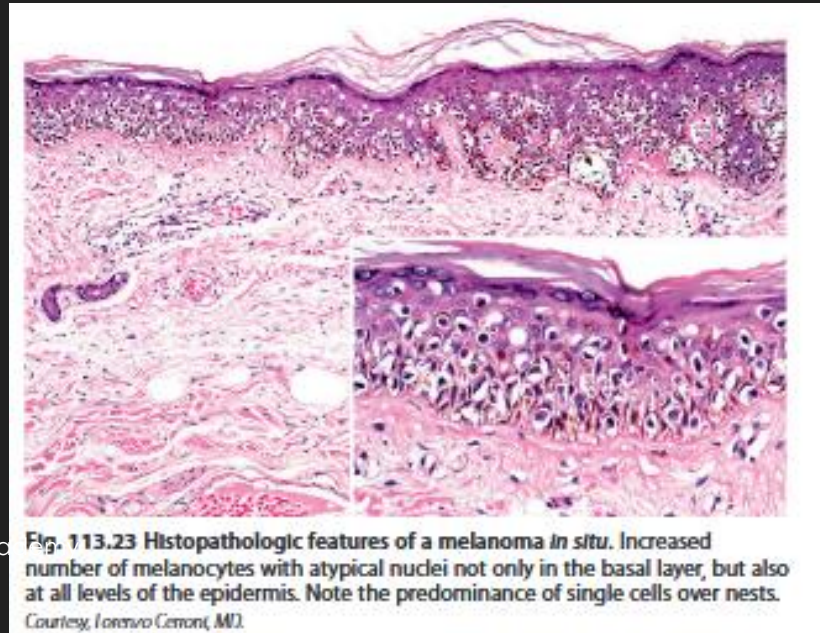


**Fig. 113.11 Melanoma *in situ*.**  
**A** Note the asymmetry, border irregularity, color variation, and erythema ("Little Red Riding Hood" sign). **B** The dermoscopic findings include a broadened network, dots and some globules inside this asymmetric lesion.



# histopathology

- The gold standard for melanoma diagnosis
- General feature :
  - Asymmetric poorly circumscribed atypical melanocyte
  - Arranged solitary or in nest



Melanoma cells  
Rounded or spindle  
shape  
Hyperchromatic  
irregular large nuclei

# Phases of growth

## *Radial growth phase*

Solitary > nest above dermoepidermal junction

3 stages :

1. Melanoma insitu
2. Micriinvasive melanoma
3. Invasive melanoma

2 pattern of melanoma cell in epidermis

1. Lentigenous pattern
2. Pagetoid spread(buckshot scatter)

## ***Vertical growth pattern***

**Progressive descent into dermis Without maturation**

**2 types**

**1- RGP COMPARTMENT** (ssm, Imm , ALM )

**2-No RGP compartment** (nodular melanoma )



# Specific histopathological finding

## SSM

Epidermal atrophy  
Pagetoid spread  
Solitary >>> up-epidermis

Perivascular inflame  
infiltirate

Nodular melanoa  
Tumor extend  
vertical in dermis

Limited or no  
epidermis

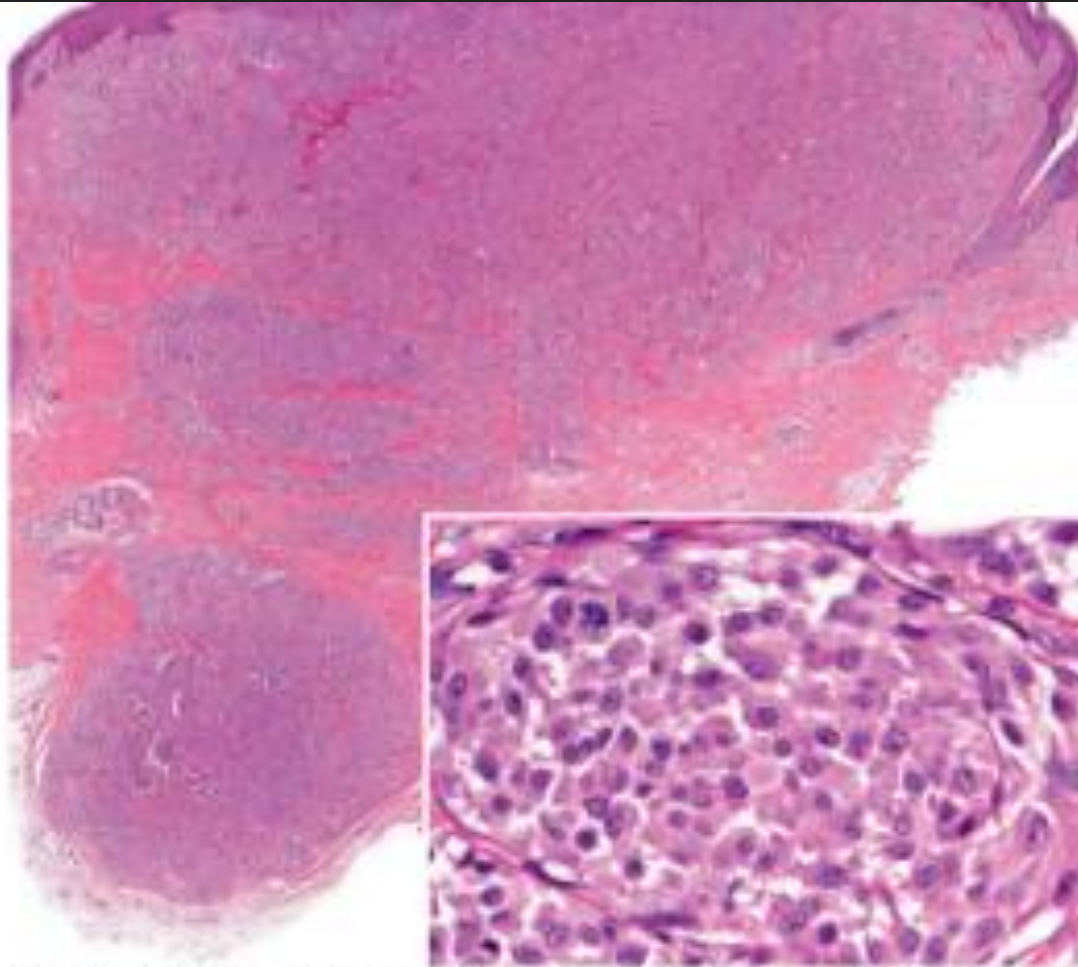
No maturation

Lintigo maligna  
Epidermal  
atrophy  
Lentigenous  
spread  
Solar elastosis in  
up- dermis

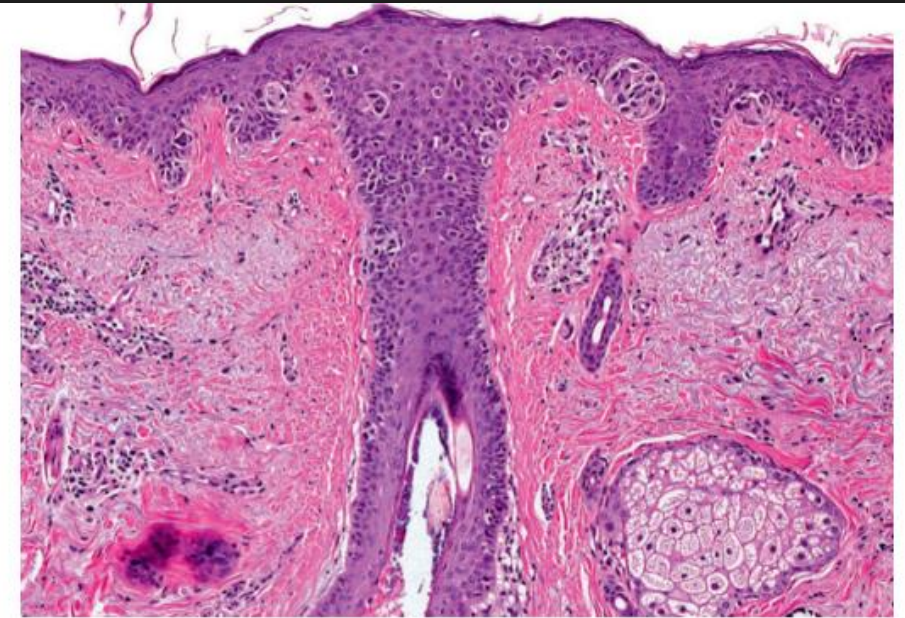
Acral lentiginous  
Hyperplastic  
epidermis

Lentigenou ,  
pagetoid spread

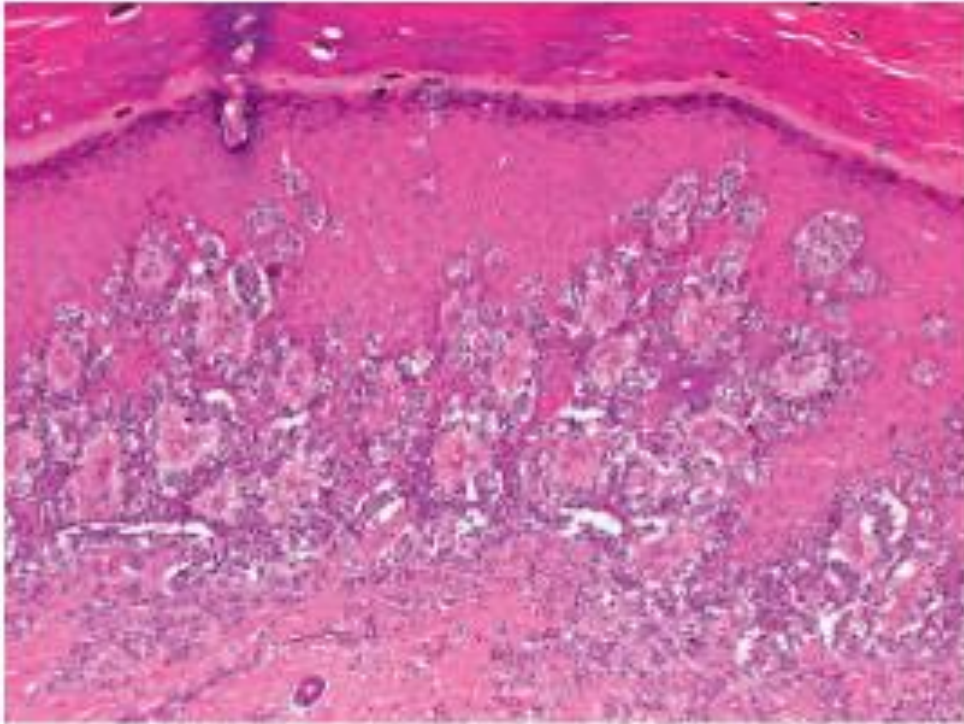




**Fig. 113.24** Histopathologic features of a nodular melanoma. Relatively well-circumscribed tumor primarily within the dermis. Confluent complexes of atypical melanocytes extend throughout the entire dermis and into the superficial portion of the subcutaneous fat. Note the atypical epithelioid melanocytes and one mitosis (inset). Courtesy, Lorenzo Cerroni, MD.



**Fig. 113.25** Histopathologic features of a lentigo maligna (melanoma *in situ* of sun-damaged skin). Proliferation of atypical melanocytes within the epidermis, with single cells predominating over nests. Note the extension of the atypical melanocytes into the hair follicle epithelium and the solar elastosis in the dermis. Courtesy, Lorenzo Cerroni, MD.



**Fig. 113.26** Histopathologic features of an acral lentiginous melanoma. Irregular, sometimes confluent, nests of atypical melanocytes within a hyperplastic epidermis. Some of the melanocytes are hyperchromatic and are also present within the cornified layer. Note the thick stratum corneum typical of acral skin. *Courtesy, Ferruccio Cerroni, MD.*

## CRITERIA FOR THE HISTOPATHOLOGIC DIAGNOSIS OF MELANOMA

### Architectural pattern

- Asymmetry
- Poor circumscription of intraepidermal melanocytic component
- Silhouette of tumor base is uneven (except in nevoid melanoma)
- No maturation of melanocytes with progressive descent into the dermis
- Nests of melanocytes within the epidermis are not equidistant from one another
- Nests of melanocytes vary in size and shape
- Some nests of melanocytes become confluent
- Scatter of melanocytes above the dermal-epidermal junction
- Melanocytes arranged as solitary units predominate over nests within the epidermis
- Melanocytes in some nests are not cohesive
- Melanocytes extend down adnexal epithelium
- Sheets of melanocytes within the dermis
- Nests at the base of lesions are occasionally large

### Cytomorphology

- Atypical melanocytes (with pleomorphic nuclei)
- Mitotic figures
- Necrotic melanocytes

### Other features

- Signs of regression
- Actinic elastosis
- "Dusty" melanin within tumor cells
- Melanin is not distributed in a uniform fashion
- Plasma cells at the base of the lesion

## CUTANEOUS MELANOMA – COMPONENTS OF THE HISTOPATHOLOGIC REPORT

### Mandatory

Diagnosis of malignant melanoma  
Tumor thickness (Breslow depth)  
Ulceration (if present)  
Margins

### Desirable

Histopathologic subtype of melanoma  
Regression, tumor-infiltrating lymphocytes, presence of plasma cells  
Vascular invasion  
Microscopic satellites  
Associated nevus  
Mitotic rate

**Table 113.7** Cutaneous melanoma – components of the histopathologic report.

# Immunohistochemistry

1. S100 :highly sensitive
2. HMB45: highly specific
3. MART 1 (MELAN A)

# Histeopathological variant of melanoma ?

# Tumor thickness

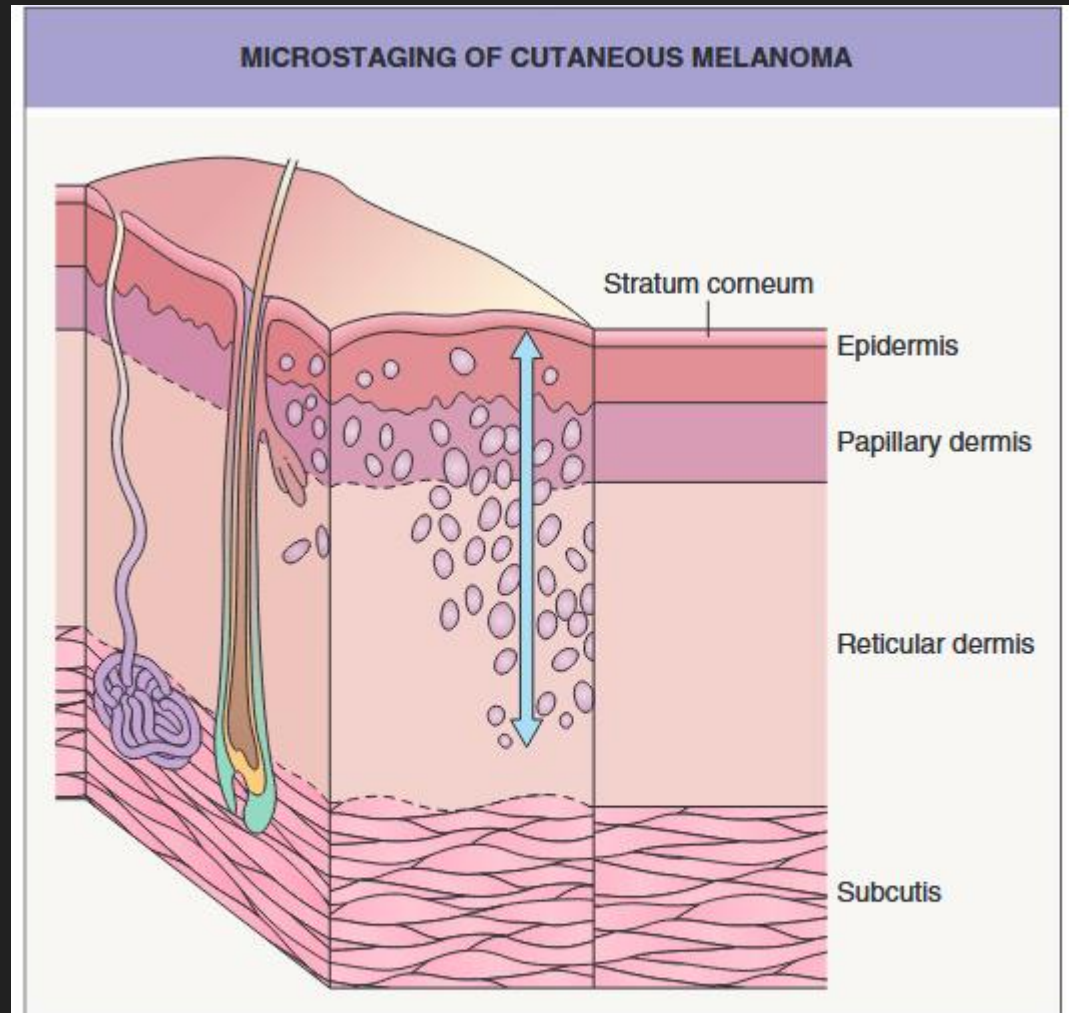
## Oclark levels:

1. Level 1: confined to epidermis
2. Level 2 : intopapillary dermis
3. Level 3:papillar – reticulated dermis interface
4. Level 4 :intosubcutaneous fat



## ○ BRESLOW INDEX

- Measurement distance between granular layer to deepest level of invasion of primary lesion
- The strongest predictor of survival



**Fig. 113.27 Microstaging of cutaneous melanoma: Breslow tumor thickness.** Breslow method: measure from the granular layer of the epidermis to the deepest part of the tumor.

## Surgical wider excision margins for primary melanoma

| Breslow thickness | Lateral excision margins to muscle or muscle fascia    |
|-------------------|--------------------------------------------------------|
| In situ           | 5-mm margins to achieve complete histological excision |
| < 1 mm            | 1 cm                                                   |
| 1.01–2 mm         | 1–2 cm                                                 |
| 2.1–4 mm          | 2–3 cm                                                 |
| > 4 mm            | 3 cm                                                   |

# AJCC Melanoma of the Skin Staging 8<sup>th</sup> Edition

## Definitions

### Primary Tumor (T)

- TX** Primary tumor cannot be assessed (for example, curettaged or severely regressed melanoma)
- T0** No evidence of primary tumor
- Tis** Melanoma in situ
- T1** Melanomas 1.0 mm or less in thickness
- T2** Melanomas 1.1 - 2.0 mm
- T3** Melanomas 2.1 - 4.0 mm
- T4** Melanomas more than 4.0 mm

**NOTE:** a and b subcategories of T are assigned based on ulceration and thickness as shown below:

| T CLASSIFICATION | THICKNESS (mm) | ULCERATION STATUS                                                                                     |
|------------------|----------------|-------------------------------------------------------------------------------------------------------|
| <b>T1</b>        | ≤1.0           | a: Breslow < 0.8 mm w/o ulceration<br>b: Breslow 0.8-1.0 mm w/o ulceration or ≤ 1.0 mm w/ ulceration. |
| <b>T2</b>        | 1.1-2.0        | a: w/o ulceration<br>b: w/ ulceration                                                                 |
| <b>T3</b>        | 2.1-4.0        | a: w/o ulceration<br>b: w/ ulceration                                                                 |
| <b>T4</b>        | >4.0           | a: w/o ulceration<br>b: w/ ulceration                                                                 |

### Regional Lymph Nodes (N)

- NX** Patients in whom the regional nodes cannot be assessed (for example previously removed for another reason)
- N0** No regional metastases detected
- N1-3** Regional metastases based on the number of metastatic nodes, number of palpable metastatic nodes on clinical exam, and presence or absence of MSI<sup>2</sup>

**NOTE:** N1-3 and a-c subcategories assigned as shown below:

| N CLASSIFICATION | # NODES   | CLINICAL DETECTABILITY/MSI STATUS                                                                                                                                                                                                  |
|------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>N1</b>        | 0-1 node  | a: clinically occult <sup>1</sup> , no MSI <sup>2</sup><br>b: clinically detected <sup>1</sup> , no MSI <sup>2</sup><br>c: 0 nodes, MSI present <sup>2</sup>                                                                       |
| <b>N2</b>        | 1-3 nodes | a: 2-3 nodes clinically occult <sup>1</sup> , no MSI <sup>2</sup><br>b: 2-3 nodes clinically detected <sup>1</sup> , no MSI <sup>2</sup><br>c: 1 node clinical or occult <sup>1</sup> , MSI present <sup>2</sup>                   |
| <b>N3</b>        | >1 nodes  | a: >3 nodes, all clinically occult <sup>1</sup> , no MSI <sup>2</sup><br>b: >3 nodes, ≥1 clinically detected <sup>1</sup> or matted, no MSI <sup>2</sup><br>c: >1 nodes clinical or occult <sup>1</sup> , MSI present <sup>2</sup> |

### Distant Metastasis (M)

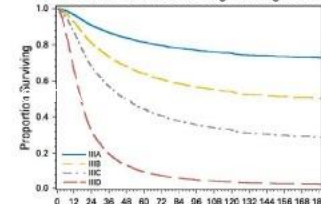
- M0** No detectable evidence of distant metastases
- M1a** Metastases to skin, sub cutaneous, or distant lymph nodes
- M1b** Metastases to lung
- M1c** Metastases to all other visceral sites
- M1d** Metastases to brain

**NOTE:** Serum LDH is incorporated into the M category as shown below:

| M CLASSIFICATION | SITE                                                                 | Serum LDH    |
|------------------|----------------------------------------------------------------------|--------------|
| <b>M1a-d</b>     | Skin/subcutaneous/nodule (a), lung (b) other visceral (c), brain (d) | Not assessed |
| <b>M1a-d(0)</b>  | Skin/subcutaneous/nodule (a), lung (b) other visceral (c), brain (d) | Normal       |
| <b>M1a-d(1)</b>  | Skin/subcutaneous/nodule (a), lung (b) other visceral (c), brain (d) | Elevated     |

| ANATOMIC STAGE/PROGNOSTIC GROUPS |       |       |    |      |        |                                 |    |      |        |        |    |
|----------------------------------|-------|-------|----|------|--------|---------------------------------|----|------|--------|--------|----|
| Clinical Staging <sup>3</sup>    |       |       |    |      |        | Pathologic Staging <sup>4</sup> |    |      |        |        |    |
| Stage 0                          | Tis   | N0    | M0 | 0    | Tis    | N0                              | M0 | 0    | Tis    | N0     | M0 |
| Stage IA                         | T1a   | N0    | M0 | IA   | T1a    | N0                              | M0 | IA   | T1a    | N0     | M0 |
| Stage IB                         | T1b   | ..    | .. | IB   | T1b    | ..                              | .. | IB   | T1b    | ..     | .. |
|                                  | T2a   | ..    | .. |      | T2a    | ..                              | .. |      | T2a    | ..     | .. |
| Stage IIA                        | T2b   | N0    | M0 | IIA  | T2b    | M0                              | M0 | IIA  | T2b    | M0     | M0 |
|                                  | T3a   | ..    | .. |      | T3a    | ..                              | .. |      | T3a    | ..     | .. |
| Stage IIB                        | T3b   | ..    | .. | IIB  | T3b    | ..                              | .. | IIB  | T3b    | ..     | .. |
|                                  | T4a   | ..    | .. |      | T4a    | ..                              | .. |      | T4a    | ..     | .. |
| Stage IIC                        | T4b   | ..    | .. | IIC  | T4b    | ..                              | .. | IIC  | T4b    | ..     | .. |
| Stage III                        | Any T | ≥N1   | M0 | IIIA | T1-2a  | N1a                             | M0 | IIIA | T1-2a  | N1a    | M0 |
|                                  | ..    | ..    | .. |      | T1-2a  | N2a                             | .. |      | T1-2a  | N2a    | .. |
|                                  | ..    | ..    | .. |      | T0     | N1b-c                           | M0 |      | T0     | N1b-c  | M0 |
|                                  | ..    | ..    | .. |      | T1-2a  | N1b-c                           | .. |      | T1-2a  | N1b-c  | .. |
|                                  | ..    | ..    | .. |      | T1-2a  | N2b                             | .. |      | T1-2a  | N2b    | .. |
|                                  | ..    | ..    | .. |      | T2b-3a | N1a-2b                          | .. |      | T2b-3a | N1a-2b | .. |
|                                  | ..    | ..    | .. |      | T0     | N2b-c                           | M0 |      | T0     | N2b-c  | M0 |
|                                  | ..    | ..    | .. |      | T0     | N3b-c                           | .. |      | T0     | N3b-c  | .. |
|                                  | ..    | ..    | .. |      | T1a-3a | N2c-3c                          | .. |      | T1a-3a | N2c-3c | .. |
|                                  | ..    | ..    | .. |      | T3b-4a | Any N                           | .. |      | T3b-4a | Any N  | .. |
|                                  | ..    | ..    | .. |      | T4b    | N1a-2c                          | .. |      | T4b    | N1a-2c | .. |
|                                  | ..    | ..    | .. |      | T4b    | N3a-c                           | M0 |      | T4b    | N3a-c  | M0 |
| Stage IV                         | Any N | Any N | M1 | IV   | Any T  | Any N                           | M1 | IV   | Any T  | Any N  | M1 |

Baseline survival after Stage III diagnosis<sup>5</sup>



Conditional, 5-year survival (%)<sup>5</sup>

| Stage | Baseline | 3y survivors |
|-------|----------|--------------|
| IIIA  | 81.4     | 83.1         |
| IIIB  | 64.0     | 76.0         |
| IIIC  | 44.5     | 66.7         |
| IIID  | 9.8      | 40.6         |

## Notes

<sup>1</sup>Nodes are designated as 'clinically detectable' if they can be palpated on physical exam and are confirmed melanoma by pathology following excision/biopsy.

<sup>2</sup>MSI comprise any satellite, locally recurrent, or in transit lesions.

<sup>3</sup>Clinical staging includes microstaging of the primary melanoma and clinical/radiologic evaluation for metastases. By convention it should be used after complete excision of the primary melanoma with clinical assessment for regional and distant metastases.

<sup>4</sup>Pathologic staging includes microstaging of the primary melanoma and pathologic information about the regional lymph nodes after partial or complete lymphadenectomy. Pathologic Stage 0 and I patients are the exceptions; they do not necessarily require pathologic evaluation of their lymph nodes. Physicians should 'discuss and consider' SLNB for patients with T1b Stage IA disease; physicians should 'discuss and offer' SLNB for patients with Stage IB disease.

# ulceration

- Absence of intact epidermis overlying major portion of tumor
- It is bad prognostic marker

# TNM STAGING

## TUMOR

T1:  $\leq 1\text{MM}$

T2 :1-2

T3: 2-4

T4:>4

a:WITHOUT ULCERATION

B: with ulceration

## L.N

N0 : NO

N1 :1INVOLVED LN

N2 :2-3 nodes

N3 :. >4

A: occult , no metastasis

B:

Detected no metastasis

C: metastasis

## METASTASIS

M0 :no

m1 :yes

M1a :skin , mucosal , non  
regional LN

M1b :lung

M1c :non CNS

M1d :CNS

# STAGES

Stage 1-2 :localized ,no nodal, no metastasis

Stage 3 :regional metastasisi, draige Ln

Stag 4 :distant metastasis



# AJCC MELANOMA TNM CLASSIFICATION – 2017

## Definition of primary tumor (T)

| T category                                                    | Thickness             | Ulceration status                             |
|---------------------------------------------------------------|-----------------------|-----------------------------------------------|
| TX (primary tumor thickness cannot be assessed <sup>†</sup> ) | NA                    | NA                                            |
| T0 (no evidence of primary tumor <sup>††</sup> )              | NA                    | NA                                            |
| Tis (melanoma in situ)                                        | NA                    | NA                                            |
| T1                                                            | ≤1.0 mm               | Unknown or unspecified                        |
| T1a                                                           | <0.8 mm               | Without ulceration                            |
| T1b                                                           | <0.8 mm<br>0.8–1.0 mm | With ulceration<br>With or without ulceration |
| T2                                                            | >1.0–2.0 mm           | Unknown or unspecified                        |
| T2a                                                           | >1.0–2.0 mm           | Without ulceration                            |
| T2b                                                           | >1.0–2.0 mm           | With ulceration                               |
| T3                                                            | >2.0–4.0 mm           | Unknown or unspecified                        |
| T3a                                                           | >2.0–4.0 mm           | Without ulceration                            |
| T3b                                                           | >2.0–4.0 mm           | With ulceration                               |
| T4                                                            | >4.0 mm               | Unknown or unspecified                        |
| T4a                                                           | >4.0 mm               | Without ulceration                            |
| T4b                                                           | >4.0 mm               | With ulceration                               |

## Definition of regional lymph node (N)

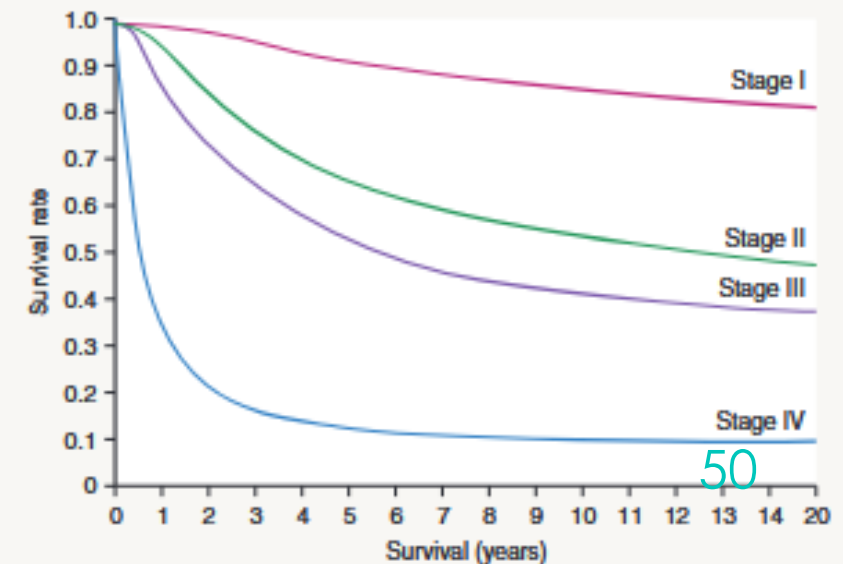
| N category | Extent of regional lymph node and/or lymphatic metastasis                                                                                                                                                                                        |                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|            | Number of tumor-involved regional lymph nodes                                                                                                                                                                                                    | Presence of in-transit, satellite, and/or microsatellite metastases* |
| NX         | Regional nodes not assessed <sup>†††</sup>                                                                                                                                                                                                       | No                                                                   |
| N0         | No regional metastases detected                                                                                                                                                                                                                  | No                                                                   |
| N1         | One tumor-involved node – or – in-transit, satellite, and/or microsatellite metastases with no tumor-involved nodes                                                                                                                              |                                                                      |
| N1a        | One clinically occult (i.e. detected by SLN biopsy)                                                                                                                                                                                              | No                                                                   |
| N1b        | One clinically detected                                                                                                                                                                                                                          | No                                                                   |
| N1c        | No regional lymph node disease                                                                                                                                                                                                                   | Yes                                                                  |
| N2         | Two or three tumor-involved nodes – or – in-transit, satellite, and/or microsatellite metastases with one tumor-involved node                                                                                                                    |                                                                      |
| N2a        | Two or three clinically occult (i.e. detected by SLN biopsy)                                                                                                                                                                                     | No                                                                   |
| N2b        | Two or three, at least one of which was clinically detected                                                                                                                                                                                      | No                                                                   |
| N2c        | One clinically occult or clinically detected                                                                                                                                                                                                     | Yes                                                                  |
| N3         | Four or more tumor-involved nodes – or – in-transit, satellite, and/or microsatellite metastases with two or more tumor-involved nodes – or – any number of matted nodes without or with in-transit, satellite, and/or microsatellite metastases |                                                                      |
| N3a        | Four or more clinically occult (i.e. detected by SLN biopsy)                                                                                                                                                                                     | No                                                                   |
| N3b        | Four or more, at least one of which was clinically detected, or presence of any number of matted nodes                                                                                                                                           | No                                                                   |
| N3c        | Two or more clinically occult or clinically detected and/or presence of any number of matted nodes                                                                                                                                               | Yes                                                                  |

# AJCC MELANOMA TNM CLASSIFICATION – 2017

## Definition of distant metastasis (M)

| M category | M criteria                                                                               |                             |
|------------|------------------------------------------------------------------------------------------|-----------------------------|
|            | Anatomic site                                                                            | LDH level <sup>4,5</sup>    |
| M0         | No evidence of distant metastasis                                                        | NA                          |
| M1         | Evidence of distant metastases                                                           |                             |
| M1a        | Distant metastasis to skin, soft tissue including muscle, and/or nonregional lymph node  | Not recorded or unspecified |
| M1a(0)     |                                                                                          | Not elevated                |
| M1a(1)     |                                                                                          | Elevated                    |
| M1b        | Distant metastasis to lung with or without M1a sites of disease                          | Not recorded or unspecified |
| M1b(0)     |                                                                                          | Not elevated                |
| M1b(1)     |                                                                                          | Elevated                    |
| M1c        | Distant metastasis to non-CNS visceral sites with or without M1a or M1b sites of disease | Not recorded or unspecified |
| M1c(0)     |                                                                                          | Not elevated                |
| M1c(1)     |                                                                                          | Elevated                    |
| M1d        | Distant metastasis to CNS with or without M1a, M1b or M1c sites of disease               | Not recorded or unspecified |
| M1d(0)     |                                                                                          | Not elevated                |
| M1d(1)     |                                                                                          | Elevated                    |

## COMPARISON OF SURVIVAL CURVES IN FOUR STAGES OF MELANOMA



# AJCC STAGE GROUPINGS FOR CUTANEOUS MELANOMA – 2017

|      | Survival (%) <sup>*</sup> | Clinical staging <sup>†</sup> |       |    | Pathologic staging <sup>‡</sup> |                      |    |
|------|---------------------------|-------------------------------|-------|----|---------------------------------|----------------------|----|
|      |                           | T                             | N     | M  | T                               | N                    | M  |
| 0    |                           | Tis                           | N0    | M0 | Tis                             | N0                   | M0 |
| IA   | 97                        | T1a                           | N0    | M0 | T1a                             | N0                   | M0 |
|      |                           |                               |       |    | T1b                             |                      |    |
| IB   | 93                        | T1b                           | N0    | M0 | T2a                             | N0                   | M0 |
|      |                           | T2a                           |       |    |                                 |                      |    |
| IIA  | 82                        | T2b                           | N0    | M0 | T2b                             | N0                   | M0 |
|      | 79                        | T3a                           |       |    | T3a                             |                      |    |
| IIB  | 68                        | T3b                           | N0    | M0 | T3b                             | N0                   | M0 |
|      | 71                        | T4a                           |       |    | T4a                             |                      |    |
| IIC  | 53                        | T4b                           | N0    | M0 | T4b                             | N0                   | M0 |
| IIIS |                           | Any T, Tis                    | ≥N1   | M0 |                                 |                      |    |
| IIIB |                           |                               |       |    | T0                              | N1b, N1c             | M0 |
| IIIC |                           |                               |       |    | T0                              | N2b, N2c, N3b or N3c | M0 |
| IIIA | 78                        |                               |       |    | T1a/b–T2a                       | N1a or N2a           | M0 |
| IIIB | 59                        |                               |       |    | T1a/b–T2a                       | N1b/c or N2b         | M0 |
|      |                           |                               |       |    | T2b/T3a                         | N1a–N2b              |    |
| IIIC | 40                        |                               |       |    | T1a–T3a                         | N2c or N3a/b/c       | M0 |
|      |                           |                               |       |    | T3b/T4a                         | Any N ≥ N1           |    |
|      |                           |                               |       |    | T4b                             | N1a–N2c              |    |
| IIID |                           |                               |       |    | T4b                             | N3a/b/c              | M0 |
| IV   | 9–27 <sup>¶</sup>         | Any T                         | Any N | M1 | Any T, Tis                      | Any N                | M1 |

## MELANOMA – MAJOR INDEPENDENT PROGNOSTIC FACTORS FOR SURVIVAL IN MULTIVARIATE ANALYSES

| Prognostic factor                | Commentary                                                                                                                    |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Tumor thickness                  | ≤1 mm, low risk; >1 mm, higher risk                                                                                           |
| Ulceration                       | Worse prognosis with ulceration                                                                                               |
| Mitotic rate                     | Worse prognosis with ≥1 mitoses/mm <sup>2</sup>                                                                               |
| Age                              | Worse prognosis with older age                                                                                                |
| Sex                              | Men have poorer prognosis (only for localized disease)                                                                        |
| Anatomic site                    | Trunk, head and neck associated with poorer prognosis than extremities                                                        |
| Number of involved lymph nodes   | Cut-off points: 1, 2–3, 4 or more lymph nodes (see Table 113.9)                                                               |
| Regional lymph node tumor burden | Clinically detectable (palpable) nodal metastases with poorer prognosis than clinically occult (microscopic) nodal metastases |
| Site of distant metastases       | Visceral metastases associated with poorer prognosis than non-visceral metastases (skin, subcutaneous, distant lymph nodes)   |

# Evaluation

1. History
2. Skin biopsy
3. Laboratory :
  1. Serum marker :protein S100B , melanoma inhibitor activity
4. Radiological :CT LUNG , MRI , PET CT

Prevention :  
Early detection :  
1. ABCD  
2. EFG  
3. CARBS

# TREATMENT

## Stag 1-2

Excision with safty margin 2cm

## Stag 3

Surgical excision with safty margin

Surgical excision of LN

Systemic therapy

Post adjuvant therapy

Apilimumab , INF alpha

## Stag 4

Surgical

Paliative radiotherapy

Systemic therapy

# treatment

surgical

systemic

BIOLOGIC  
BRAF INHIBITOR  
MEK inh.

Immunotherapy  
antiCTLA4  
antiPD1  
IL-2

Chemotherapy  
melphalan



| TARGETED THERAPIES AND CHECKPOINT INHIBITORS  |               |                                                                  |
|-----------------------------------------------|---------------|------------------------------------------------------------------|
| Targeting the MAPK pathway (see Fig. 113.2)   |               | FDA approved as of July 2017                                     |
| BRAF inhibitors (selective*)                  | Dabrafenib    | ✓                                                                |
|                                               | Encorafenib   |                                                                  |
|                                               | Vemurafenib   | ✓                                                                |
| MEK inhibitors                                | Binimetinib   |                                                                  |
|                                               | Cobimetinib   | ✓                                                                |
|                                               | Trametinib    | ✓                                                                |
| Immune checkpoint inhibitors (see Fig. 128.9) |               |                                                                  |
| Anti-CTLA-4 antibody                          | Ipilimumab    | ✓                                                                |
|                                               | Tremelimumab  |                                                                  |
| Anti-PD-1 antibody                            | Nivolumab     | ✓                                                                |
|                                               | Pembrolizumab | ✓                                                                |
|                                               | Pidilizumab   |                                                                  |
| Anti-PD-L1 antibodies                         | Atezolizumab  | Approved for non-small cell lung cancer and urothelial carcinoma |
|                                               | Avelumab      | Approved for metastatic Merkel cell carcinoma                    |
|                                               | Durvalumab    | Approved for bladder cancer (PD-L1-positive)                     |

| Immunotherapy    | MOA                                                       | S/E                                                                                                                                                                                                                                                                                                                    |
|------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ipilimumab       | blocks a protein on the surface of T-cells, called CTLA-4 | dermatitis, colitis, hepatitis, uveitis, neuropathy, and endocrine dysfunction. Reactions may be mild but occasionally life-threatening                                                                                                                                                                                |
| Nivolumab        | binds to PD1 protein                                      | diarrhoea, skin reactions, tiredness, feeling sick, loss of appetite, liver upset, changes in gland function (e.g. thyroid and adrenals), joint pain, and eye inflammation. These side-effects are the same as the side-effects of ipilimumab, but are less common than with ipilimumab, and less likely to be severe. |
| Pembrolizumab    | binds to PD1 protein                                      | diarrhoea, skin reactions, tiredness, feeling sick, loss of appetite, liver upset, changes in gland function (e.g. thyroid and adrenals), joint pain, and eye inflammation. These side-effects are the same as the side-effects of ipilimumab, but are less common than with ipilimumab, and less likely to be severe. |
| Targeted Therapy | MOA                                                       |                                                                                                                                                                                                                                                                                                                        |
| Vemurafenib      | Blocks BRAF V60                                           | joint pain, tiredness, acneiform rashes, high temperature, photosensitivity, ECG changes, liver dysfunction, nausea, hair loss, itching and the development of new AKs and SCCs                                                                                                                                        |
| Dabrafenib       | Blocks BRAF V60                                           | decreased appetite, high temperature, headache, cough, nausea, vomiting, liver dysfunction, diarrhoea, rashes, ECG changes, joint pain and hair loss.                                                                                                                                                                  |
| Trametinib       | Blocks abnormal BRAF protein activity (MEK inhibitor)     | tiredness, nausea, changes in eyesight, weakness in heart muscle, headache, chills, diarrhoea, rash, joint pain, high blood pressure and vomiting.                                                                                                                                                                     |

| CHEMOTHERAPY DRUGS FOR METASTATIC MELANOMA |                                                                                                                                              |                           |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Medication                                 | Dose                                                                                                                                         | Objective response rates* |
| Dacarbazine (DTIC), 5 days                 | 250 mg/m <sup>2</sup> IV daily for 5 days every 3–4 weeks                                                                                    | 12–17%                    |
| Dacarbazine (DTIC), 1 day                  | 800–1200 mg/m <sup>2</sup> IV on one day every 3–4 weeks                                                                                     | 5–23%                     |
| Temozolomide                               | 150–200 mg/m <sup>2</sup> po daily for 5 days every 4 weeks                                                                                  | 13–21%                    |
| Fotemustine                                | 100 mg/m <sup>2</sup> IV on days 1, 8 and 15; then 5-week pause, then repeat single dose every 3 weeks                                       | 7–24%                     |
| Vindesine                                  | 3 mg/m <sup>2</sup> IV every 14 days                                                                                                         | 12–26%                    |
| Interferon- $\alpha$                       | 9–18 million IU/m <sup>2</sup> sc 3x weekly                                                                                                  | (13–25%)                  |
| Interleukin-2                              | 600 000 IU/kg as 15-min infusion IV every 8 hours for 5 days (total of 14 doses). Repeat every 2 weeks, depending upon response and toxicity | (16–21%)                  |

